

Table 2. Data Validation Summary

Validation item	Value	Interpretation
Mean missing ratio	0.1967	Across all session-level missingness summaries
Worst session missing ratio	0.4559	Highest session-level missingness
Minimum session missing ratio	0.0672	Lowest session-level missingness
Total timestamp gaps detected	0	Gap threshold from validation output
Largest temporal gap	0.00 ms	Maximum detected temporal gap
Mean median frame interval	40.5 ms	Average of per-session median intervals
Main outlier family	motion speed and selected keypoint coordinates (top examples: p2_motion_speed, p2_kpt50_world_y, p2_kpt50_world_x)	Z-score and IQR outlier checks
Interpretation	Usable dataset, but validity coverage must be considered	Missingness/visibility features remain analytically relevant

Validation summary for missingness, temporal consistency, and extreme outliers.